

Task No.: 5**Date: 08-01-2026**

Title: Develop an application that makes use of Notification Manager**Aim:**

To create an Android application that makes use of the Notification Manager in the application.

Code:*MainActivity.kt*

```
package com.example.madtask6
```

```
import android.Manifest
import android.app.NotificationChannel
import android.app.NotificationManager
import android.app.PendingIntent
import android.content.Context
import android.content.Intent
import android.content.pm.PackageManager
import android.net.Uri
import android.os.Build
import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.app.NotificationCompat
import androidx.core.content.ContextCompat
import androidx.core.app.Person
import java.util.Date
```

```
class MainActivity : AppCompatActivity() {
```

```
    private lateinit var notificationManager: NotificationManager
    private val CHANNEL_ID = "notify_channel"
```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    // 🛎 Runtime notification permission (Android 13+)
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        if (ContextCompat.checkSelfPermission(
            this,
            Manifest.permission.POST_NOTIFICATIONS
        ) != PackageManager.PERMISSION_GRANTED
        ) {
            ActivityCompat.requestPermissions(
                this,
                arrayOf(Manifest.permission.POST_NOTIFICATIONS),
                100
            )
        }
    }

    notificationManager =
        getSystemService(Context.NOTIFICATION_SERVICE) as NotificationManager

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        val channel = NotificationChannel(
            CHANNEL_ID,
            "Notification Manager",
            NotificationManager.IMPORTANCE_HIGH
        )
        notificationManager.createNotificationChannel(channel)
    }

    findViewById<Button>(R.id.btnSimple).setOnClickListener { simpleNotification() }
    findViewById<Button>(R.id.btnIcon).setOnClickListener { notificationWithIcon() }
    findViewById<Button>(R.id.btnImage).setOnClickListener { notificationWithImage() }
    findViewById<Button>(R.id.btnGroup).setOnClickListener { notificationWithGroup() }
}

    findViewById<Button>(R.id.btnAction).setOnClickListener { notificationSemantic() }
}

private fun simpleNotification() {
    val builder = NotificationCompat.Builder(this, CHANNEL_ID)

```

```

        .setSmallIcon(R.mipmap.ic_launcher)
        .setContentTitle("Simple Notification")
        .setContentText("This is a simple notification")

notificationManager.notify(1, builder.build())
}

private fun notificationWithIcon() {
    val builder = NotificationCompat.Builder(this, CHANNEL_ID)
        .setSmallIcon(R.mipmap.ic_launcher)
        .setContentTitle("Notification with Icon")
        .setContentText("Notification includes an icon")

    notificationManager.notify(2, builder.build())
}

private fun notificationWithImage() {
    val uri = Uri.parse("android.resource://$packageName/drawable/sample_image")
    val bot = Person.Builder().setName("Bot").setBot(true).build()

    val message = NotificationCompat.MessagingStyle.Message(
        "Image Notification",
        Date().time,
        bot
    )
    message.setData("image/*", uri)

    val builder = NotificationCompat.Builder(this, CHANNEL_ID)
        .setSmallIcon(R.mipmap.ic_launcher)
        .setStyle(NotificationCompat.MessagingStyle(bot).addMessage(message))

    notificationManager.notify(3, builder.build())
}

private fun notificationWithGroup() {
    val user = Person.Builder().setName("User").build()
    val bot = Person.Builder().setName("Bot").setBot(true).build()

    val style = NotificationCompat.MessagingStyle(bot)
        .addMessage("Hello", Date().time, user)
        .addMessage("Hi!", Date().time, bot)

```

```

        .setGroupConversation(true)
        .setConversationTitle("Group Chat")

    val builder = NotificationCompat.Builder(this, CHANNEL_ID)
        .setSmallIcon(R.mipmap.ic_launcher)
        .setStyle(style)

    notificationManager.notify(4, builder.build())
}

private fun notificationSemantic() {
    val intent = Intent(this, MainActivity::class.java)
    intent.putExtra("msg", "Notification Read")

    val pendingIntent = PendingIntent.getActivity(
        this, 0, intent,
        PendingIntent.FLAG_UPDATE_CURRENT or PendingIntent.FLAG_IMMUTABLE
    )

    val action = NotificationCompat.Action.Builder(
        R.mipmap.ic_launcher,
        "MARK READ",
        pendingIntent
    )

    .setSemanticAction(NotificationCompat.Action.SEMANTIC_ACTION_MARK_AS_READ)
        .build()

    val builder = NotificationCompat.Builder(this, CHANNEL_ID)
        .setSmallIcon(R.mipmap.ic_launcher)
        .setContentTitle("Semantic Action")
        .setContentText("Tap to mark as read")
        .addAction(action)

    notificationManager.notify(5, builder.build())
}

override fun onResume() {
    super.onResume()
    intent.getStringExtra("msg")?.let {
        Toast.makeText(this, it, Toast.LENGTH_SHORT).show()
    }
}

```

```

    }
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <TextView
        android:id="@+id/txtName"
        android:text="Name: Yoga Shri P K"
        android:textSize="18sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/txtRoll"
        android:text="Roll No: 23BCS192"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginTop="4dp"
        app:layout_constraintTop_toBottomOf="@id/txtName"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/txtTitle"

```

```

android:text="Notification Manager"
android:textSize="22sp"
android:textStyle="bold"
android:layout_marginTop="20dp"
app:layout_constraintTop_toBottomOf="@id/txtRoll"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
android:layout_width="wrap_content"
android:layout_height="wrap_content"/>

```

<Button

```

android:id="@+id/btnSimple"
android:text="SIMPLE NOTIFICATION"
android:background="@drawable/button_shape"
android:textColor="@android:color/white"
android:layout_width="0dp"
android:layout_height="75dp"
android:layout_marginTop="25dp"
app:layout_constraintTop_toBottomOf="@id/txtTitle"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"/>

```

<Button

```

android:id="@+id/btnIcon"
android:text="NOTIFICATION WITH ICON"
android:background="@drawable/button_shape"
android:textColor="@android:color/white"
android:layout_width="0dp"
android:layout_height="75dp"
android:layout_marginTop="12dp"
app:layout_constraintTop_toBottomOf="@id/btnSimple"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"/>

```

<Button

```

android:id="@+id/btnImage"
android:text="NOTIFICATION WITH IMAGE"
android:background="@drawable/button_shape"

```

```

android:textColor="@android:color/white"
android:layout_width="0dp"
android:layout_height="75dp"
android:layout_marginTop="12dp"
app:layout_constraintTop_toBottomOf="@id/btnIcon"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"/>

```

<Button

```

android:id="@+id/btnGroup"
android:text="NOTIFICATION OF GROUP CONVERSATION"
android:background="@drawable/button_shape"
android:textColor="@android:color/white"
android:layout_width="0dp"
android:layout_height="100dp"
android:layout_marginTop="12dp"
app:layout_constraintTop_toBottomOf="@id/btnImage"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"/>

```

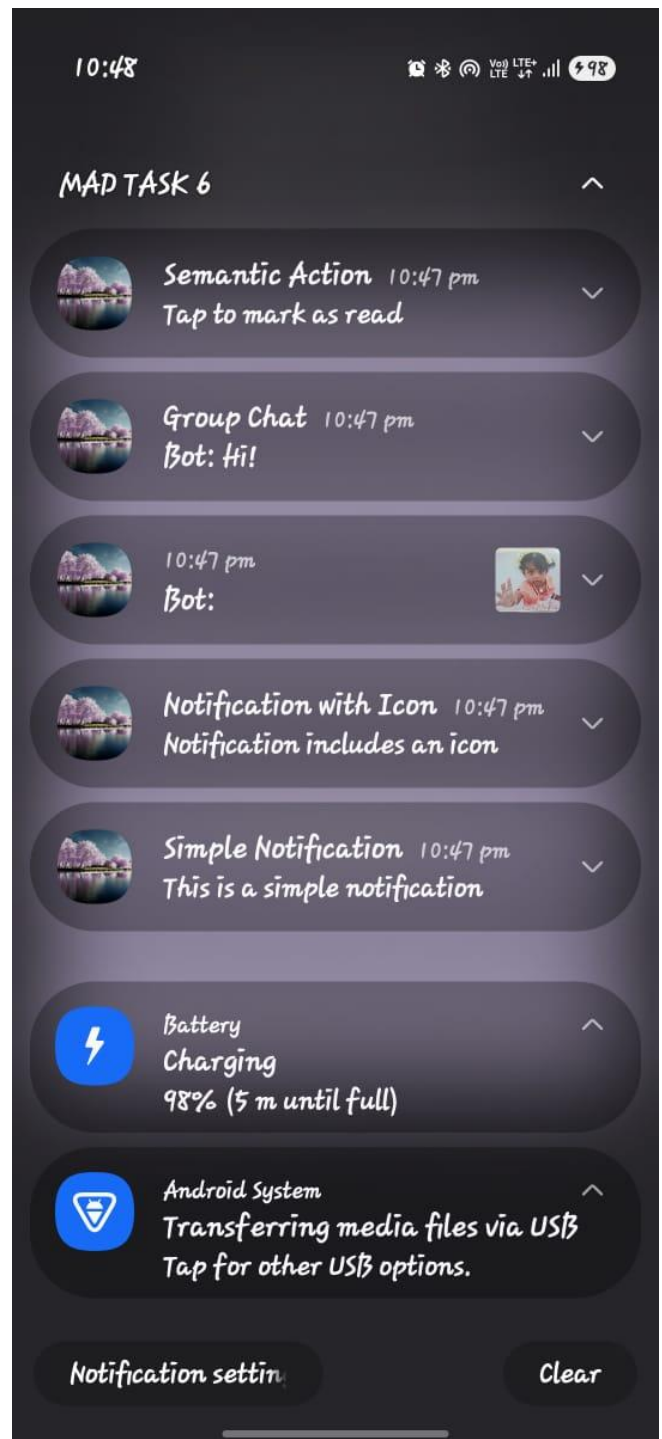
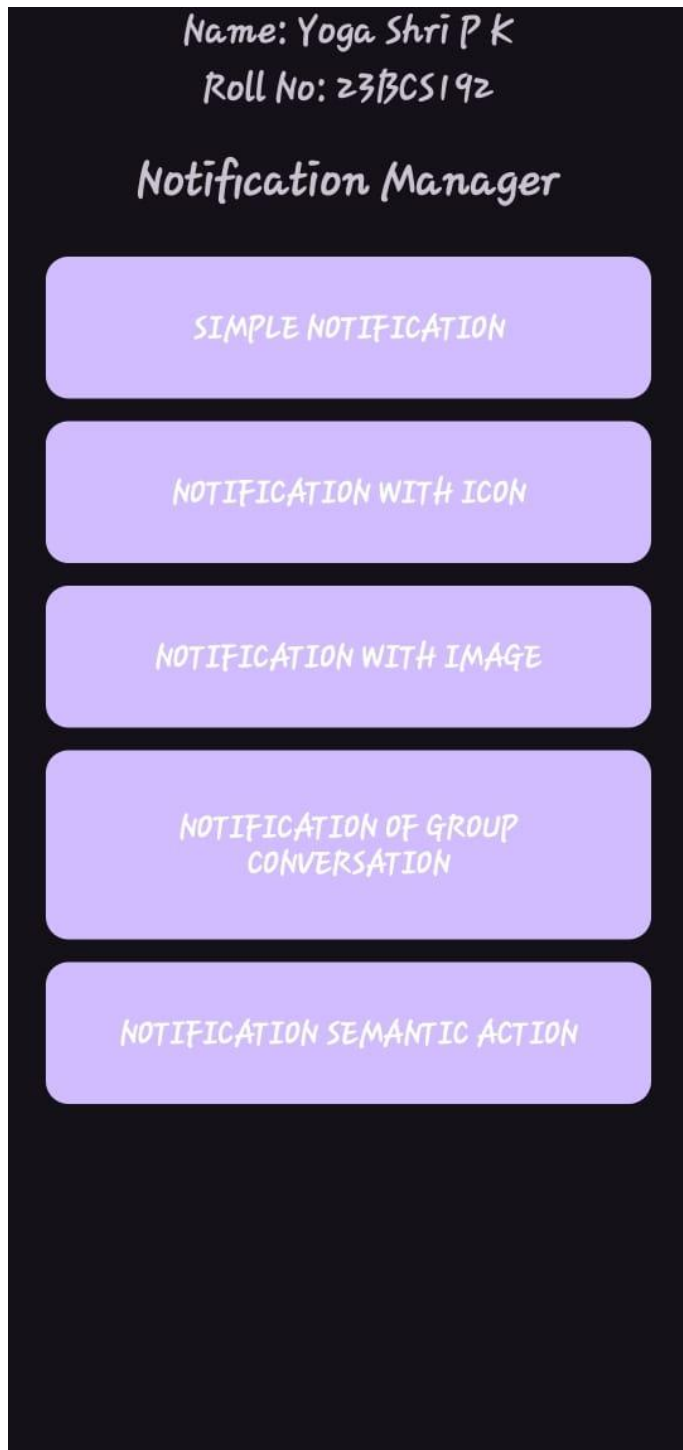
<Button

```

android:id="@+id/btnAction"
android:text="NOTIFICATION SEMANTIC ACTION"
android:background="@drawable/button_shape"
android:textColor="@android:color/white"
android:layout_width="0dp"
android:layout_height="75dp"
android:layout_marginTop="12dp"
app:layout_constraintTop_toBottomOf="@id/btnGroup"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"/>

```

</androidx.constraintlayout.widget.ConstraintLayout>

Output:**Result:**

An android application using the notification manager is executed successfully.